

Total No. of Questions : 10]

SEAT No. :

P6524

[Total No. of Pages : 2

[5870]-2005

T.E. (Computer Engineering)
EMBEDDED OPERATING SYSTEMS
(2012 Pattern) (Semester - II) (310250)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Explain the RISC architectural features of ARM. [6]
b) What are the qualities of Good RTOS? [4]

OR

- Q2)** a) What is deadlock? List the necessary conditions for a deadlock to occur. How it is resolved? [6]
b) What is Readers-Writers problem? [4]

- Q3)** a) What is Busy Box? How to configure it? Explain its usefulness in embedded systems. [6]
b) What is cross development environment for Linux? [4]

OR

- Q4)** a) Name and explain different components of bootstrap loader? [6]
b) What are the main categories of kernel releases? [4]

- Q5)** a) What is core dump? How to debug a core dump? [6]
b) What are the different types of device drivers? Explain depmod and rmmod. [6]
c) Name and explain a Linux utility used for partitioning the block devices. [5]

OR

P.T.O.

- Q6)** a) Explain the following Linux utilities used: [6]
i) fdisk ii) mount iii) mkfs
b) How to build device drivers in Embedded Linux? [6]
c) What are the types of file systems used for flash memory? [5]

- Q7)** a) How do modern processors and compilers make it difficult to debug Linux kernel? [6]
b) What are binutils? Name and explain three of them. [6]
c) What is JTAG probe debugging? [5]

OR

- Q8)** a) Why tracing and profiling tools are required? Name and explain 3 such tools. [6]
b) Explain in detail Embedded Graphics Processing. [5]
c) Discuss the challenges in debugging a Linux application code. [6]

- Q9)** a) What are real-time processes? Which latency periods affect their performance? [6]
b) What are the issues involved in preempting the Linux kernel? [6]
c) Give details of bootloader for embedded Android. [4]

OR

- Q10)** a) What are the scheduling policies used by Linux to schedule real time processes? [6]
b) Explain the role of Zygote, system server in embedded android. [6]
c) What are the types of real-time systems? [4]

